This is a Title

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1 Setting up LaTeX in VScodium

2 Basic Text

This is normal text.
Bold text
Italic text
Underlined text

Centered Text

Right-aligned Text

Left-aligned Text

This is big text
This is larger text
This is large text
This is normal text
This is small text
This is footnote size text
This is script size text
This is tiny text

3 Lists

- First item
- Second item

1. Step One
2. Step Two

4 Equations

Inline maths: $E = mc^2$.

$$F = ma \tag{1}$$

Reference Equation 1.

5 Matrices

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

6 Calculus

$$\frac{\mathrm{d}y}{\mathrm{d}x}, \qquad \frac{\partial f}{\partial x}, \qquad \int_0^1 x^2 \, dx$$

7 Vectors and Physics

$$\mathbf{F} = m\mathbf{a}$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

8 Quantum Mechanics

$$|\psi\rangle, \qquad \langle\phi|, \qquad \langle\phi|\psi|\phi|\psi\rangle$$

9 Units

$$c = 2.997\,924\,58 \times 10^8 \,\mathrm{m\,s^{-1}}$$

10 Table Example

x	y
1	2
2	4

Table 1: Example Table

11 Code Example

```
import numpy as np
x = np.linspace(0,10,100)
print(x)
```

12 Figures

Add figures using:

```
\includegraphics[width=0.5\textwidth]{LaTeX/LaTeX_meme.png}
```

13 Hyperlinks

https://www.ucc.ie

14 Conclusion

This document demonstrates the most useful LaTeX features for undergraduate physics students.

References